Google Cloud H2S

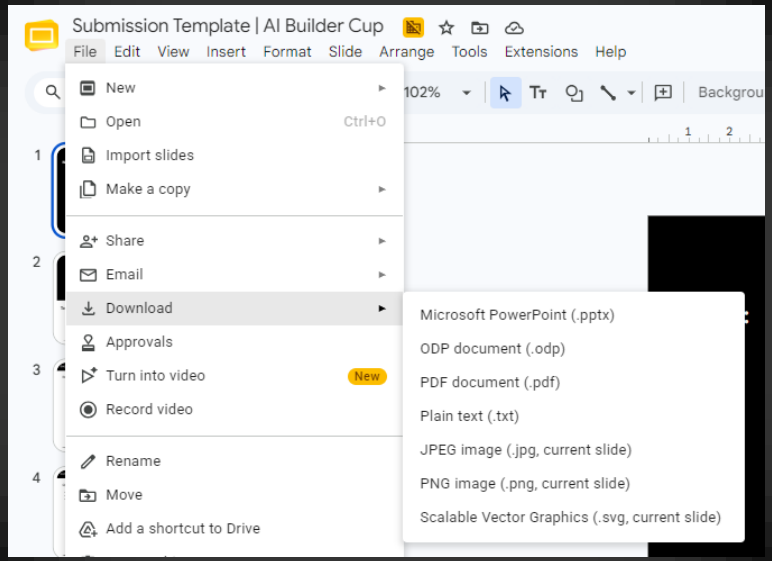
AI Builder Cup

Build in JAPAC.
Build for the world.



## Team Details

a. Team name: Scam Shield
b. Team leader name: Naman Shrimal
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences. People in India lose thousands of crores to scam messages, QR codes and calls that look real, with no quick way to check them or to turn a scam into action.




Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

# Brief about the idea

**Scam Shield checks the thing you were just sent before you act on it:** a text, screenshot, QR code, link or voice note.

## Verdict you can trust to read

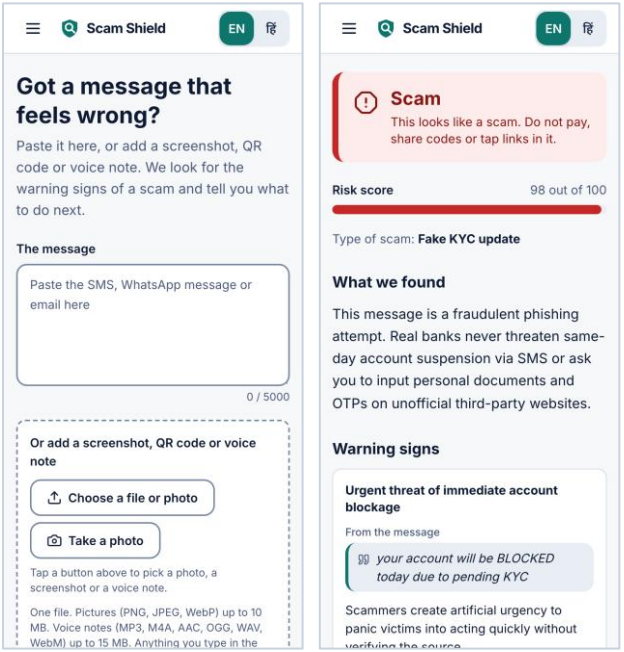
Scam, Suspicious or No scam signs found, with the red flags quoted from the message, and what to do next. In English or Hindi; Hinglish is understood.

## From a verdict to action

For a scam: a ready-to-file 1930 / cybercrime.gov.in complaint, a shared scam list so the next person is warned, and a WhatsApp warning card.

## Safe by design

Links open in an isolated browser, never on the person's phone. It never says "safe"; guidance only.




The live app: paste or share, then the verdict with evidence

# Opportunities

**Rs 19,813 crore** reported cyber-fraud losses in 2025 | about 21.8 lakh complaints | about 2% recovered (I4C data)

## How different is it?

- Truecaller Scam Checker (Sep 2026): matches links, numbers and text already reported; no screenshots, QR or voice.
- Google scam detection: mostly Pixel phones.
- Chakshu: a reporting form; it gives no verdict.
- Scam Shield analyses the content itself.

## How does it solve the problem?

- Checks before you pay, scan or click, in seconds, in Hindi or English.
- Explains every red flag with the exact words from the message.
- Turns a scam into action: complaint draft, shared scam list, family warning card.

## USP

- Catches scams nobody has reported yet.
- UPI-aware: a QR that asks you to pay while the text says receive; fake payment receipts.
- Hinglish voice notes (fake police "digital arrest" calls).
- Safe link scanning in an isolated browser.

List of features offered by the solution

### 5 input types
Text, screenshot, QR code, link and voice note.

### Clear verdict
Scam, Suspicious or No scam signs found, plus a risk score. Never "safe".

### Evidence
Every red flag quotes the words or part of the image it is based on.

### English + Hindi
Hinglish understood; answers in the language you pick.

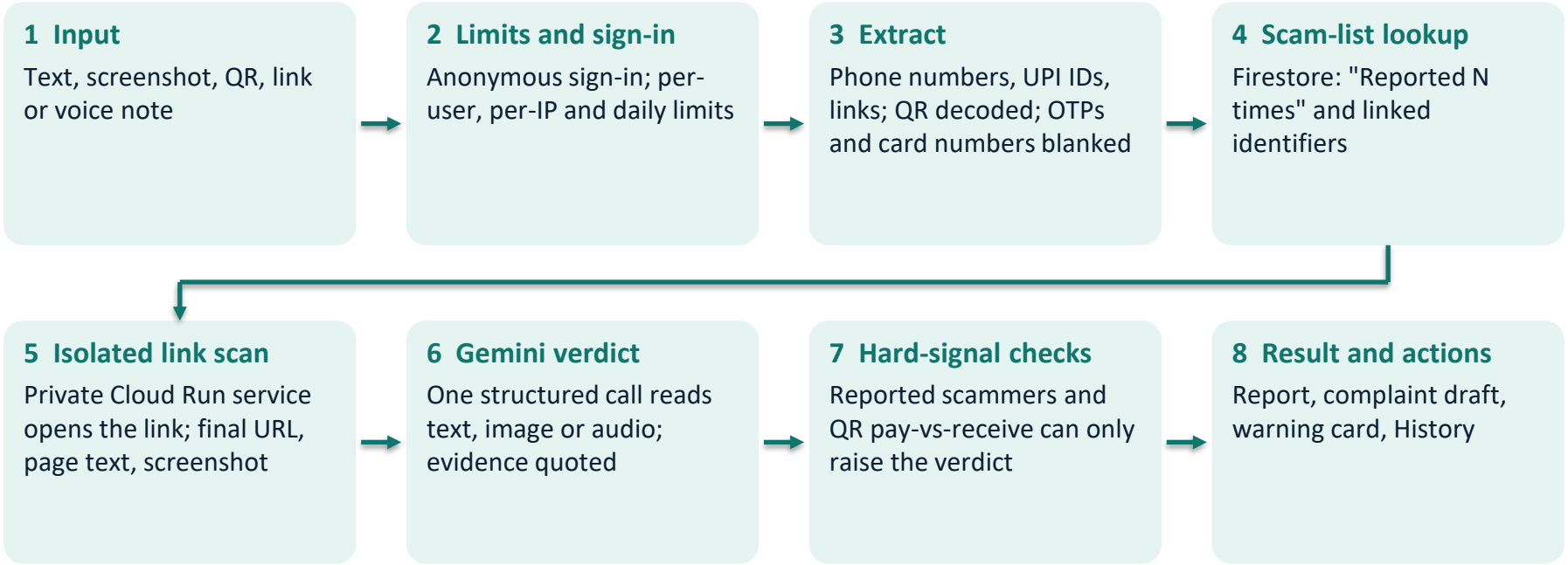
### Complaint draft
Ready-to-file text for 1930 and cybercrime.gov.in.

### Shared scam list
"Reported N times", with linked numbers, UPI IDs and websites.

### Warning card
One tap to share a warning to the family WhatsApp group.

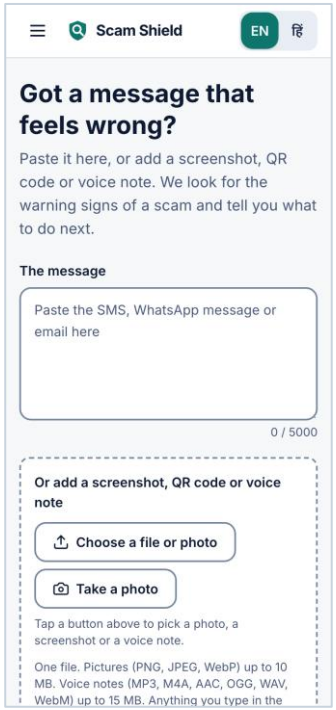
### Everyday extras
Trends, history, one-click examples, installable app, Android share menu, WebMCP agent tools.

# Process flow diagram or Use-case diagram

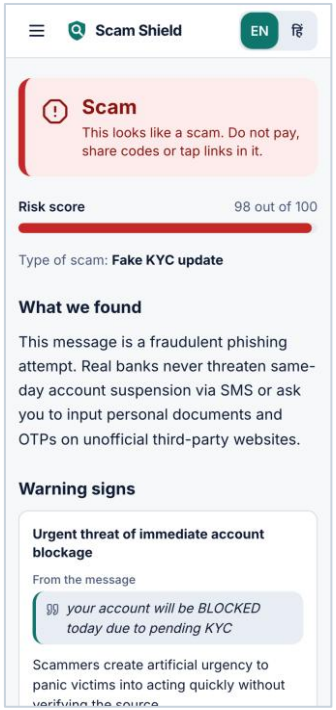



*If Gemini fails or times out, the person sees a clear retry message; no verdict is invented and nothing is stored.*

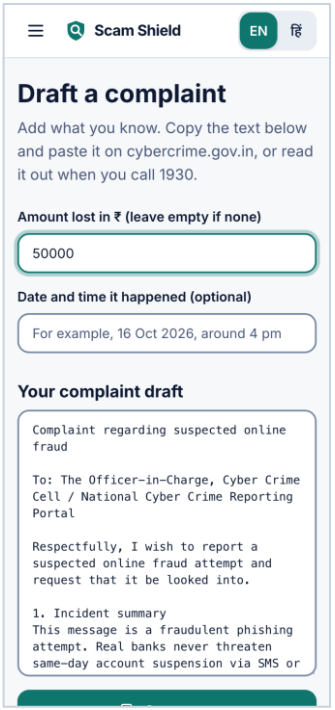
# Wireframes/Mock diagrams of the proposed solution (optional)




Check




Result

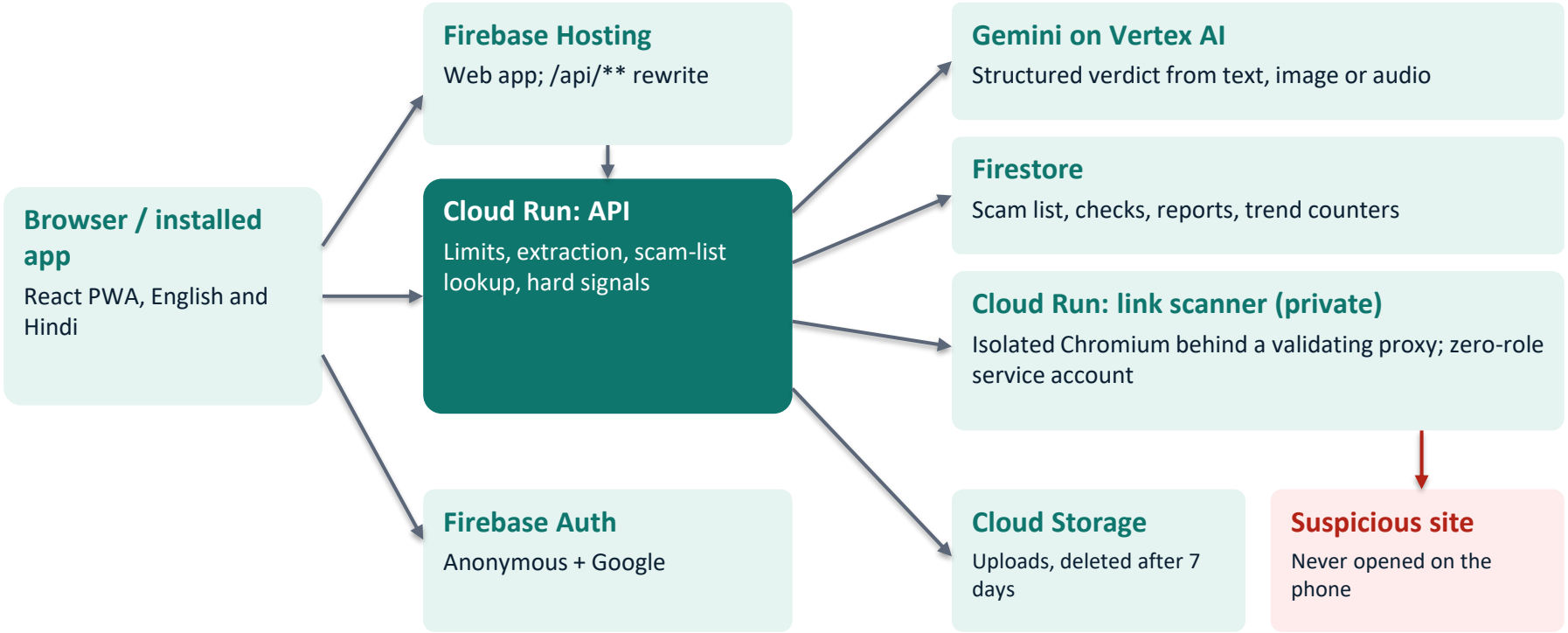



Complaint draft




Hindi

# Architecture diagram of the proposed solution




*The web app reads its own checks and the public trends from Firestore; uploads go straight to Storage.*

# Technologies to be used in the solution

## Google Cloud and Firebase

- Gemini on Vertex AI: gemini-3.7-flash, with gemini-2.5-flash as the automatic fallback; text, image and audio in one structured call
- Cloud Run: two services (public API, private link scanner)
- Firestore: scam list, checks, reports, trends, usage limits
- Firebase Authentication (anonymous, Google), Hosting, Storage
- Cloud Build and Artifact Registry for deploys; Cloud Text-to-Speech for the voice example

## Application

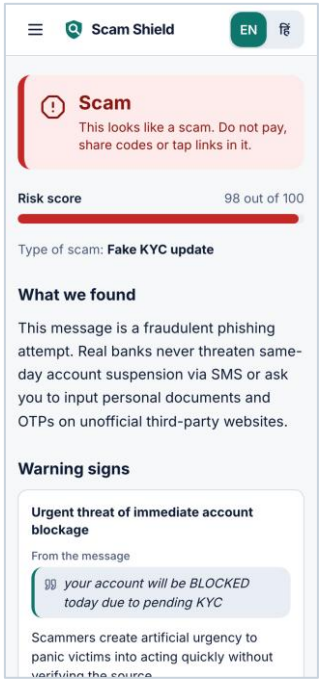
- React 18, Vite, Tailwind, Workbox PWA with an Android share target
- Node 22, TypeScript, Hono on Cloud Run
- Playwright / Chromium for the isolated link scanner
- zod schemas shared by web and API; sharp and jsQR for QR decoding
- Vitest: over 550 automated tests, plus a 50-case live benchmark

# Estimated implementation cost (optional)

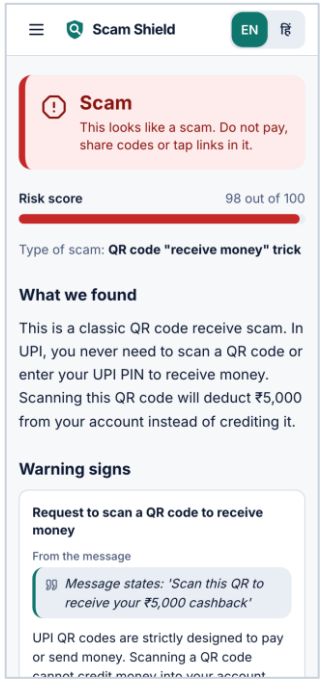
| Component | Cost driver | Note |
|---|---|---|
| Cloud Run API | One always-warm instance (1 vCPU, 1 GiB) | Main fixed cost; avoids cold starts for users and judges |
| Cloud Run link scanner | Per scan | Scales to zero when idle |
| Gemini on Vertex AI | Per check (roughly 2 to 3 thousand input and 500 output tokens) | Capped by 30 checks per user and 3,000 checks per day |
| Firebase Hosting, Auth, Firestore, Storage | Usage | Expected to stay in free tiers at this volume |
| Budget alert | Rs 1,000 per month | Alerts at 50%, 90% and 100% of the budget |

*Estimates only. Exact figures depend on traffic and current Google Cloud prices; the budget alert is set on the project's billing account.*

# Snapshots of the prototype



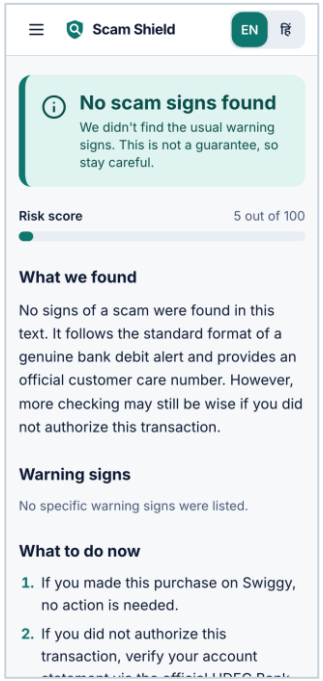

Scam verdict with evidence



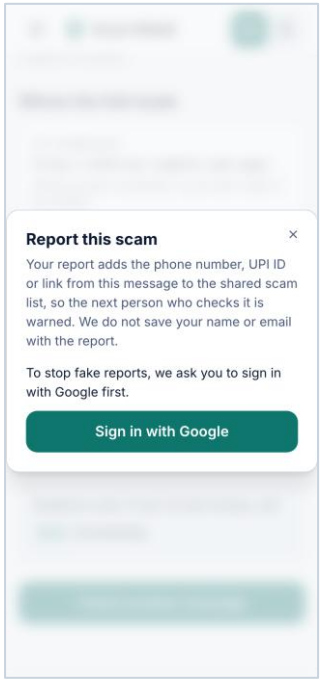

QR that asks you to pay




Hindi voice note




No scam signs found




Report to the scam list

# Prototype Performance report/Benchmarking

**100.0%**
Accuracy

**100.0%**
Scam recall

**0.0%**
False-positive rate

**96.2%**
Scam type correct

| Input | Median | Slowest 5% | Cases |
|---|---|---|---|
| Text | 4.5 s | 12.9 s | 45 |
| Image | 8.9 s | 9.5 s | 4 |
| Audio | 14.2 s | 14.2 s | 1 |

*Slowest 5% is the 95th-percentile time. Image checks vary most because Gemini latency varies.*

- 50 synthetic cases (26 scams across all 13 scam types, 24 genuine messages), run one at a time against the live app.
- Errors: 0. Genuine message flagged: none.
- The first run found a real bug (the model did not know today's date and flagged 2026 dates as future); it was fixed and re-measured.
- Model choice: an earlier run on gemini-3.8-flash scored 98.0% accuracy and 4.2% false positives, with image checks up to 34 s. It was slow and rate-limited on this project, so the default is gemini-3.7-flash with a fallback.
- Small and synthetic: a sanity check, not a guarantee.

# Additional Details/Future Development (if any)

### More Indian languages
Tamil, Telugu, Marathi, Bengali and more.

### Call recordings and on-device checks
Check a call as it happens; keep private content on the phone.

### Bank and telecom partners
Feed verified reports into the scam list and push warnings.

### URL reputation feeds
Add Google Web Risk next to the isolated scan.

### Android app
Share-sheet and notification checks for SMS and WhatsApp.

### Abuse protection
Firebase App Check and a stricter per-device limit.

### Google Search grounding
Switch on for link checks once project quota allows.

### Misinformation mode
The same engine for "is this forward real?".

Provide links to your:

- GitHub Public Repository: https://github.com/Namans12/scam-shield
- Demo Video Link (3 Minutes): [add the demo video link before submitting]
- Final Product Link: https://project-91260b43-6705-4176-b71.web.app

Google Cloud H2S

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# Thank you

